

Haskell

PENG

September 29, 2026

Contents

1	Introduction	1
1.1	Functional	1
1.2	Pure	1
1.3	Lazy	1
1.4	Statically Typed	2
1.5	Themes	2
1.6	Declarations and variables	2
1.7	Basic Types	2
1.8	Arithmetic	2
1.9	Boolean Logic	2
1.10	Defining Basic Functions	2

1 Introduction

<https://www.engineering.upenn.edu/~cis1940/spring13/>

Haskell is a *lazy, functional* programming language created in the late 1980s.

1.1 Functional

Definition 1.1 (functional).

- *functions are values*
- *evaluating expressions rather than executing instructions*

1.2 Pure

Definition 1.2 (referentially transparent).

- *everything (var, data structures) is immutable*
- *expressions have no side effects*
- *same function and arguments = same output*

benefits:

- replace equals by equals
- parallelism

1.3 Lazy

Definition 1.3 (lazy). *Haskell expressions are not evaluated until their results are actually needed.*

consequences:

- define a control structure by defining a function
- define infinite data structures
- time and space usage become more complicated

1.4 Statically Typed

Definition 1.4 (statically typed). *Each Haskell expression has a type, and types are all checked at compile-time.*

1.5 Themes

- static type system
- abstraction principle: *every idea, algorithm, and piece of data should occur exactly once in your code*
- wholemeal programming: *first solve a more general problem, then transform it into more specialized ones*

1.6 Declarations and variables

Definition 1.5 (=). *= denotes definition such that variables are not mutable*

Example 1.6. `x :: Int`

`x = 3`

`x = 4 -- error, multiple declaration of x`

`y :: Int`

`y = y + 1 -- <<loop>>`

1.7 Basic Types

- `Int` : at least up to $\pm 2^{29}$
- `Integer`
- `Double`
- `Float`
- `Bool`
- `Char`
- `String`

1.8 Arithmetic

Example 1.7.

`mod 18 5`

`18 'mod' 5`

`2 / 2 -- floating-point division`

`div 2 2 -- integer division`

`2 'div' 2`

1.9 Boolean Logic

Definition 1.8 (boolean logic). `&&`, `||`, `not`

Definition 1.9 (equality and order).

- `==` or `/=`
- `>`, `<`, `>=`, `<=`

Definition 1.10 (if-expression). `if b then t else f`

Remark 1.11. *For if-expression, the else part is required.*

1.10 Defining Basic Functions